

Multiple Linear Regression

POLS 602

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Review

1. What does $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$ mean, in words?
2. What does SSE measure, and why is it important?
3. Why might we want to think about regression in terms of conditional means?

Today

- A motivating example for multiple regression
- Defining multiple regression
- Interpreting regression coefficients

Suppose I have the following DGP:

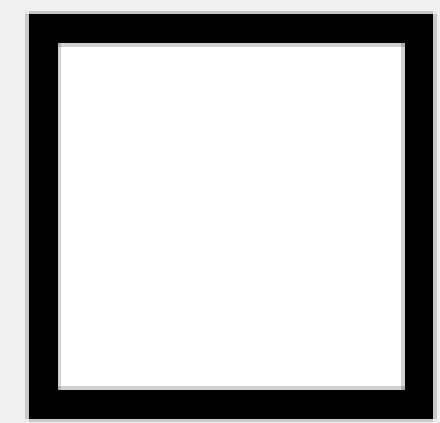
$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \epsilon_i$$

I want to estimate the effect of x_i on y_i (that is, β_1 is my estimand and $\hat{\beta}_1$ is my estimate). Which is the correct estimator?

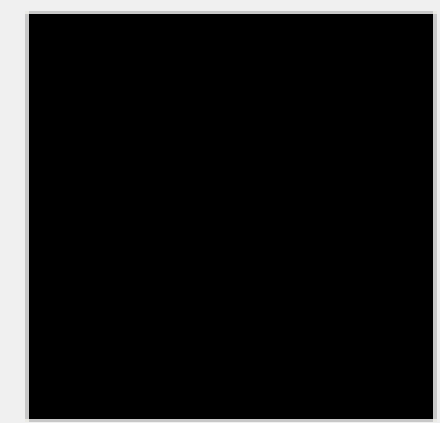
A. $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\epsilon}_i$

B. $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\beta}_2 z_i + \hat{\epsilon}_i$

Test Scores by School Type



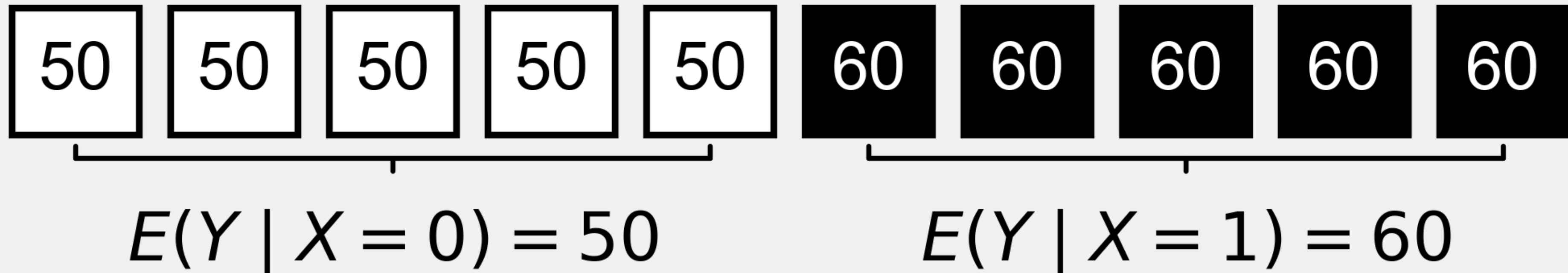
Public School



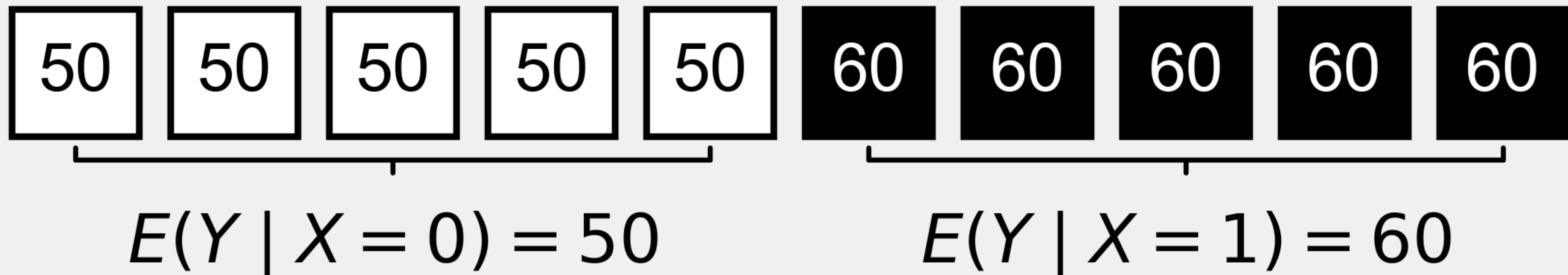
Private School

Y = Test Score, X = Private School

Test Scores by School Type



Test Scores by School Type



$$\hat{\beta}_1 = 10$$

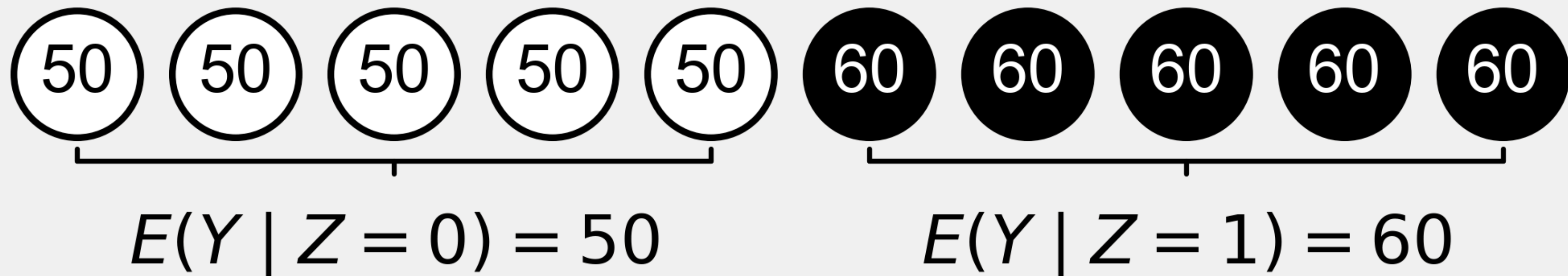
Test Scores by Wealth

○ Not Wealthy

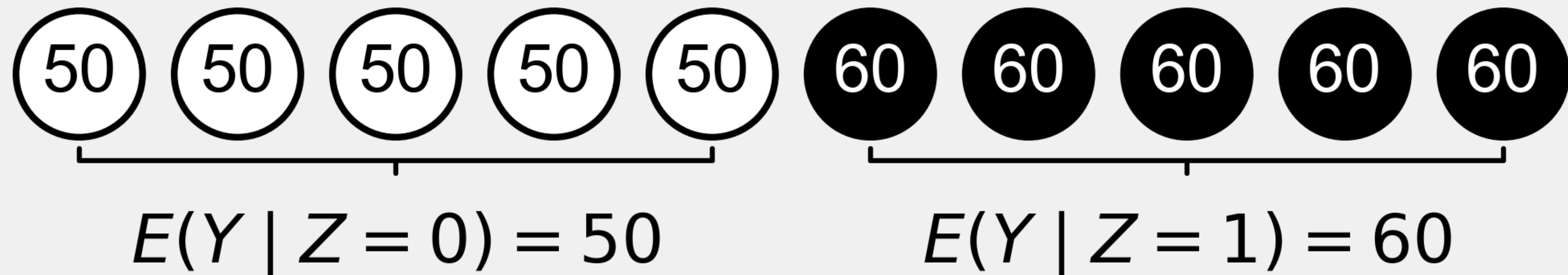
● Wealthy

Y = Test Score, Z = Wealthy

Test Scores by Wealth

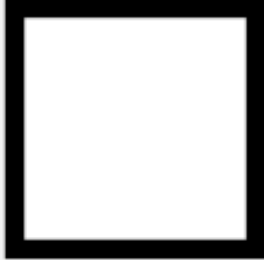


Test Scores by Wealth



$$\hat{\beta}_1 = 10$$

Combining Both Predictors

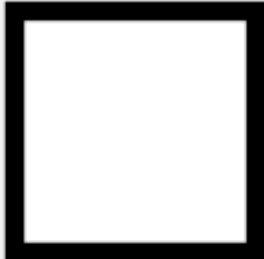
 Public School

 Private School

 Not Wealthy

 Wealthy

The True Data-Generating Process

 Public School

 Private School

 Not Wealthy

 Wealthy

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \epsilon_i$$

Y = Test Score, X = Private School, Z = Wealthy

$\beta_0 = 50, \beta_1 = 10, \beta_2 = 10$

The Observed Sample



$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \epsilon_i$$

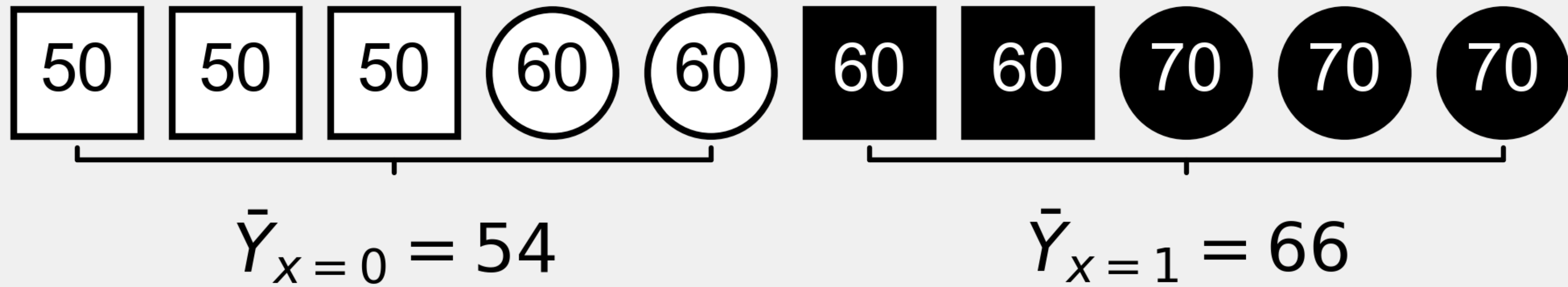
Hypothesized model



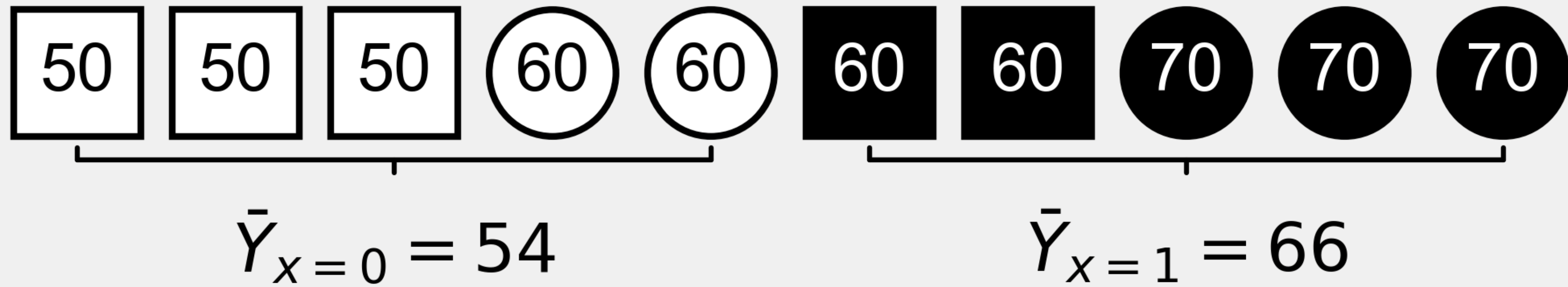
$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\epsilon}_i$$

- $x_i = 1$ if the student went to a private school
- notice the z_i variable drops

Hypothesized model



Hypothesized model



$$\hat{\beta}_1 = 12$$

Bias

A systematic tendency for an estimator to miss the true value, in the same direction, on average

- Our hypothesized model gave $\hat{\beta}_1 = 12$, but the true effect is $\beta_1 = 10$
- This is different from noise: a biased estimator doesn't average out to the truth even with more data, because the problem is what's in the model, not how much data we have
- Bias is a property of the estimator's *sampling* distribution, not any single sample.
- The type of bias in our example is known as *omitted variable bias*

Bias, Formally

$$\text{Bias}(\hat{\beta}) = E[\hat{\beta}] - \beta$$

- $E[\hat{\beta}]$: the average value $\hat{\beta}$ would take across repeated samples
- β : the true parameter value
- An estimator is **unbiased** if $\text{Bias}(\hat{\beta}) = 0$, i.e. $E[\hat{\beta}] = \beta$
- Here: $E[\hat{\beta}_1] = 12 \neq 10 = \beta_1$, so omitting z makes $\hat{\beta}_1$ **biased**

Confounding Variables

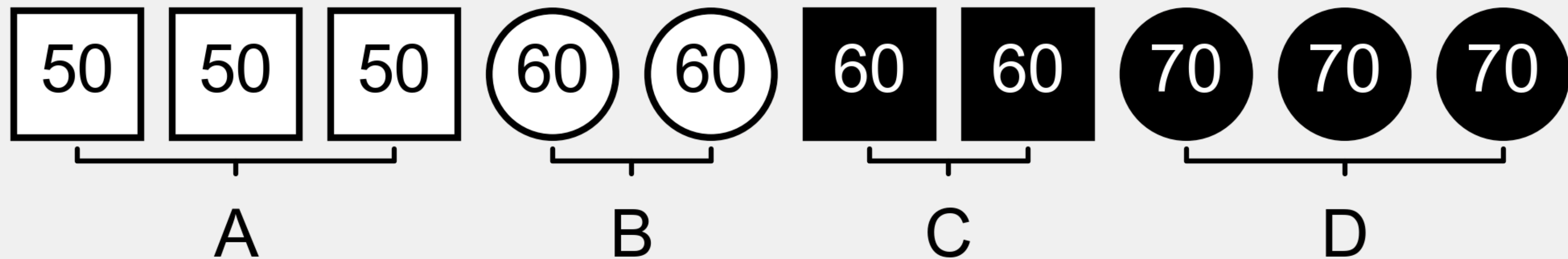
A variable that has a causal effect on both the independent and dependent variable. Confounders will bias the estimate of your independent variable if not accounted for.

Holding Wealth Constant



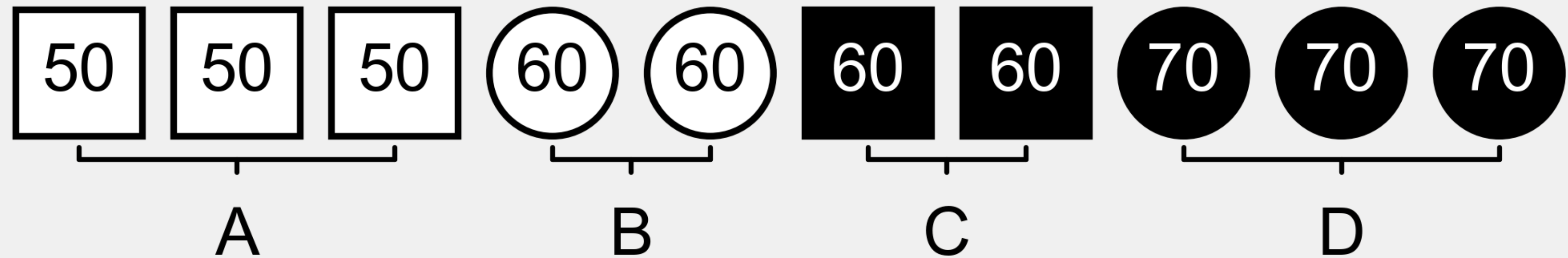
$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\beta}_2 z_i + \hat{\epsilon}_i$$

Holding Wealth Constant



$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\beta}_2 z_i + \hat{\epsilon}_i$$

Holding Wealth Constant



$$\hat{\beta}_1 \approx \frac{[\bar{C} - \bar{A}] + [\bar{D} - \bar{B}]}{2} \approx 10$$

*Note this is an approximation, not how a regression actually calculates $\hat{\beta}_1$.

Multiple Regression

The Multiple Regression Model

For k predictors $X_1, \dots, X_k$:

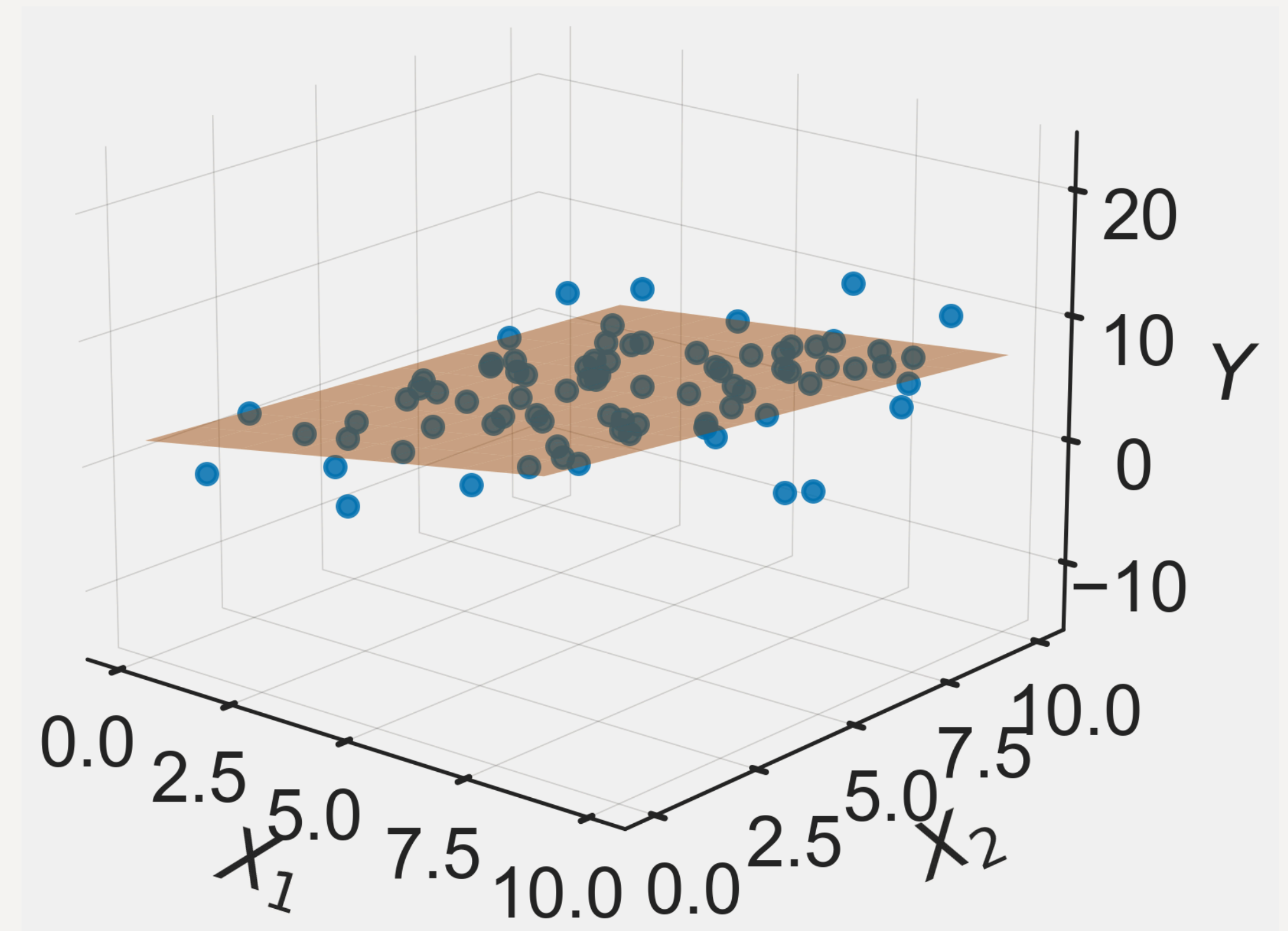
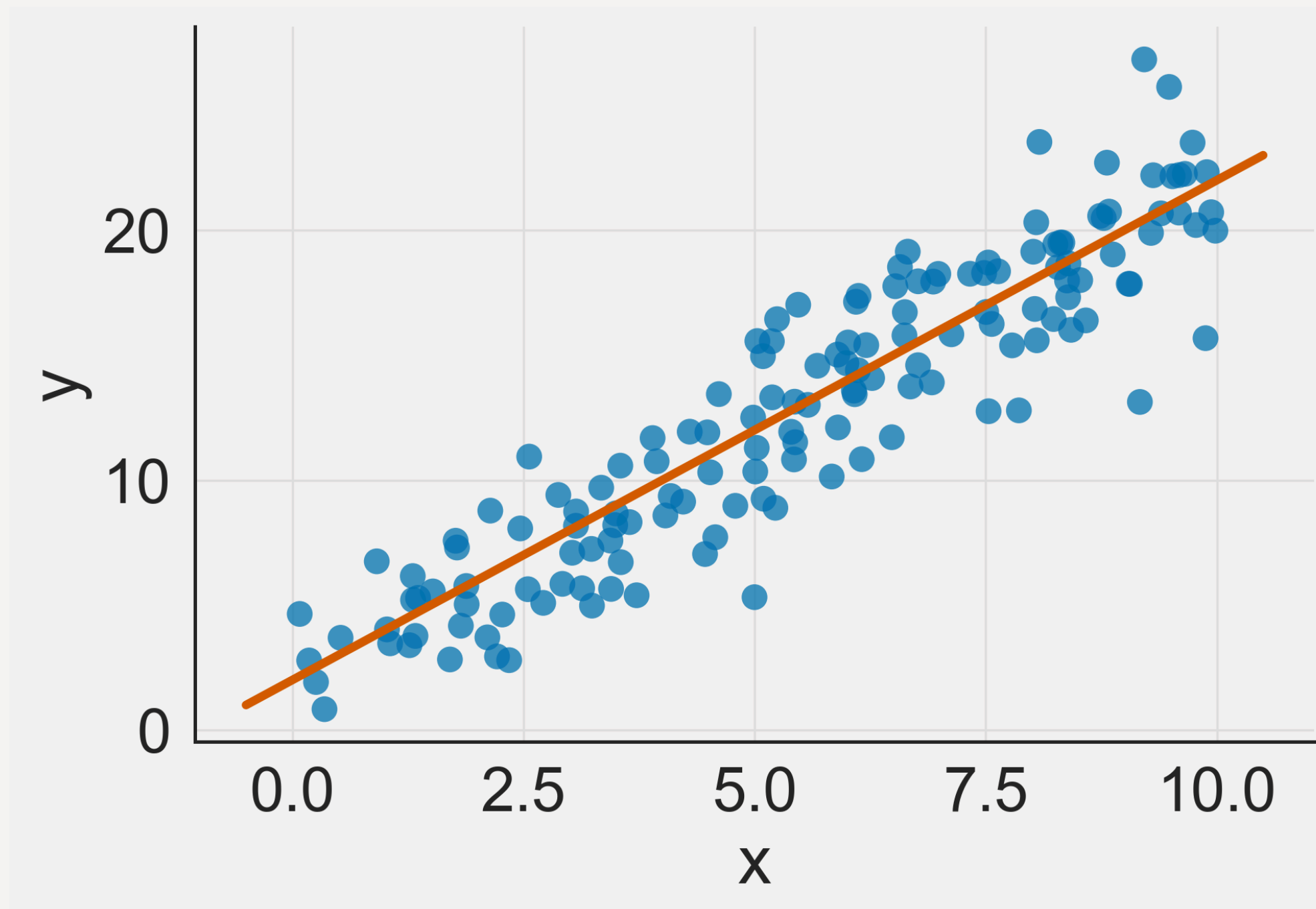
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i$$

- Same idea as simple regression, just with more terms

The Multiple Regression Model

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$$

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_{1i} + \widehat{\beta}_2 X_{2i}$$



The Multiple Regression Model

For k predictors $X_1, \dots, X_k$:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i$$

- Same idea as simple regression, just with more terms
- β_j is the effect of X_j on Y , **holding all other predictors in the model constant**. But what exactly does this mean?

Holding Variables Constant

A regression coefficient is a **derivative**.

In the bivariate case:

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$$

$$\frac{d\widehat{Y}}{dX} = (0) + \widehat{\beta}_1 (1)$$

$$\frac{d\widehat{Y}}{dX} = \widehat{\beta}_1$$

Holding Variables Constant

Now with k predictors:

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_{1i} + \widehat{\beta}_2 X_{2i} + \cdots + \widehat{\beta}_k X_{ki}$$

Differentiate with respect to X_1 , holding $X_2, \dots, X_k$ as constants:

$$\frac{\partial \widehat{Y}}{\partial X_1} = (0) + \widehat{\beta}_1 (1) + \widehat{\beta}_2 (0) + \cdots + \widehat{\beta}_k (0)$$

$$\frac{\partial \widehat{Y}}{\partial X_1} = \widehat{\beta}_1$$

Multiple Regression In Matrix Form

$$Y = X\beta + \varepsilon$$

Looks identical to the bivariate case, the difference is entirely inside **X**.

The Multiple Regression Design Matrix

For n observations and k predictors:

$$\mathbf{X} = \begin{pmatrix} 1 & X_{1,1} & X_{2,1} & \cdots & X_{k,1} \\ 1 & X_{1,2} & X_{2,2} & \cdots & X_{k,2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1,n} & X_{2,n} & \cdots & X_{k,n} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}$$

The School/Wealth Example, in Matrix Form

$$\mathbf{X} = \begin{pmatrix} 1 & \text{private}_1 & \text{wealthy}_1 \\ 1 & \text{private}_2 & \text{wealthy}_2 \\ \vdots & \vdots & \vdots \\ 1 & \text{private}_n & \text{wealthy}_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

Multiplying It Out

Compute the dot product of row i of $\mathbf{X}$ with $\boldsymbol{\beta}$:

$$\begin{aligned} & (1)(\beta_0) + (\text{private}_i)(\beta_1) + (\text{wealthy}_i)(\beta_2) \\ &= \beta_0 + \beta_1 \text{private}_i + \beta_2 \text{wealthy}_i \end{aligned}$$

Estimation Is the Same Idea

- We still estimate $\hat{\beta}$ by minimizing SSE
- $SSE = \hat{\varepsilon}^\top \hat{\varepsilon}$, where $\hat{\varepsilon} = \mathbf{Y} - \mathbf{X}\hat{\beta}$
- We'll derive the formula for $\hat{\beta}$ that minimizes SSE soon

Dummy and Categorical Variables

A Dummy Variable

- A **dummy variable** takes only two values, 0 and 1, marking group membership
- We've already used one: $\text{private}_i = 1$ if student i attends private school, 0 otherwise
- Also called an **indicator or binary variable**

Interpreting a Dummy's Coefficient

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 D_i, \quad D_i \in \{0, 1\}$$

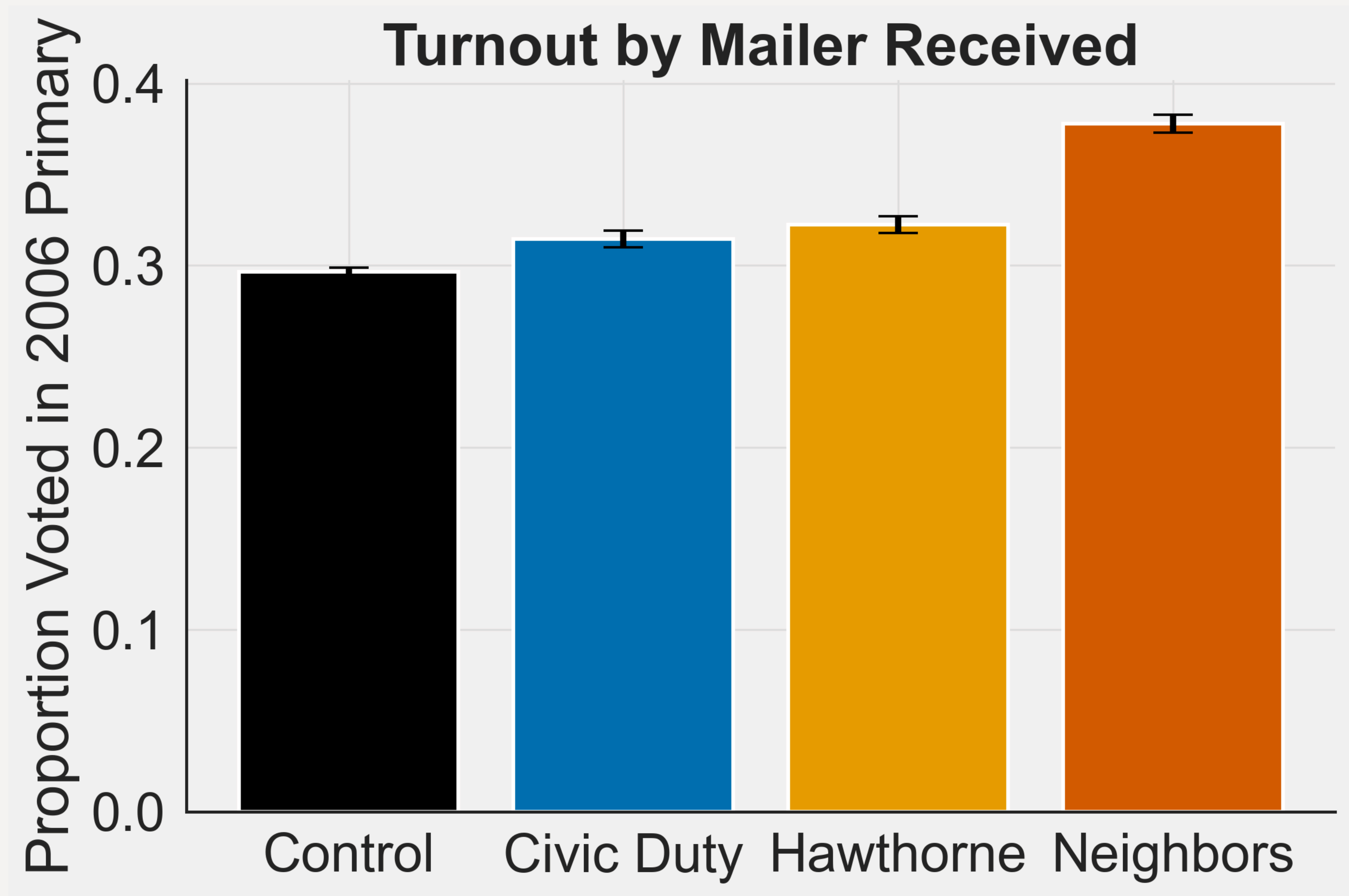
- $D_i = 0$: $\widehat{Y}_i = \widehat{\beta}_0$
- $D_i = 1$: $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1$

$\widehat{\beta}_1$ is the difference in predicted Y between the two groups.

From Two Groups to Several

- The Gerber, Green, and Larimer (2008) mailer experiment had **four** groups, not two: Control, Civic Duty, Hawthorne, Neighbors
- A single 0/1 dummy can't represent 4 categories
- Solution: use **multiple dummies**, one for each non-baseline category

Turnout by Mailer Received



The Categorical Regression Model

With Control as the baseline:

$$\begin{aligned}\widehat{Y}_i = & \widehat{\beta}_0 + \widehat{\beta}_1 \text{CivicDuty}_i \\ & + \widehat{\beta}_2 \text{Hawthorne}_i + \widehat{\beta}_3 \text{Neighbors}_i\end{aligned}$$

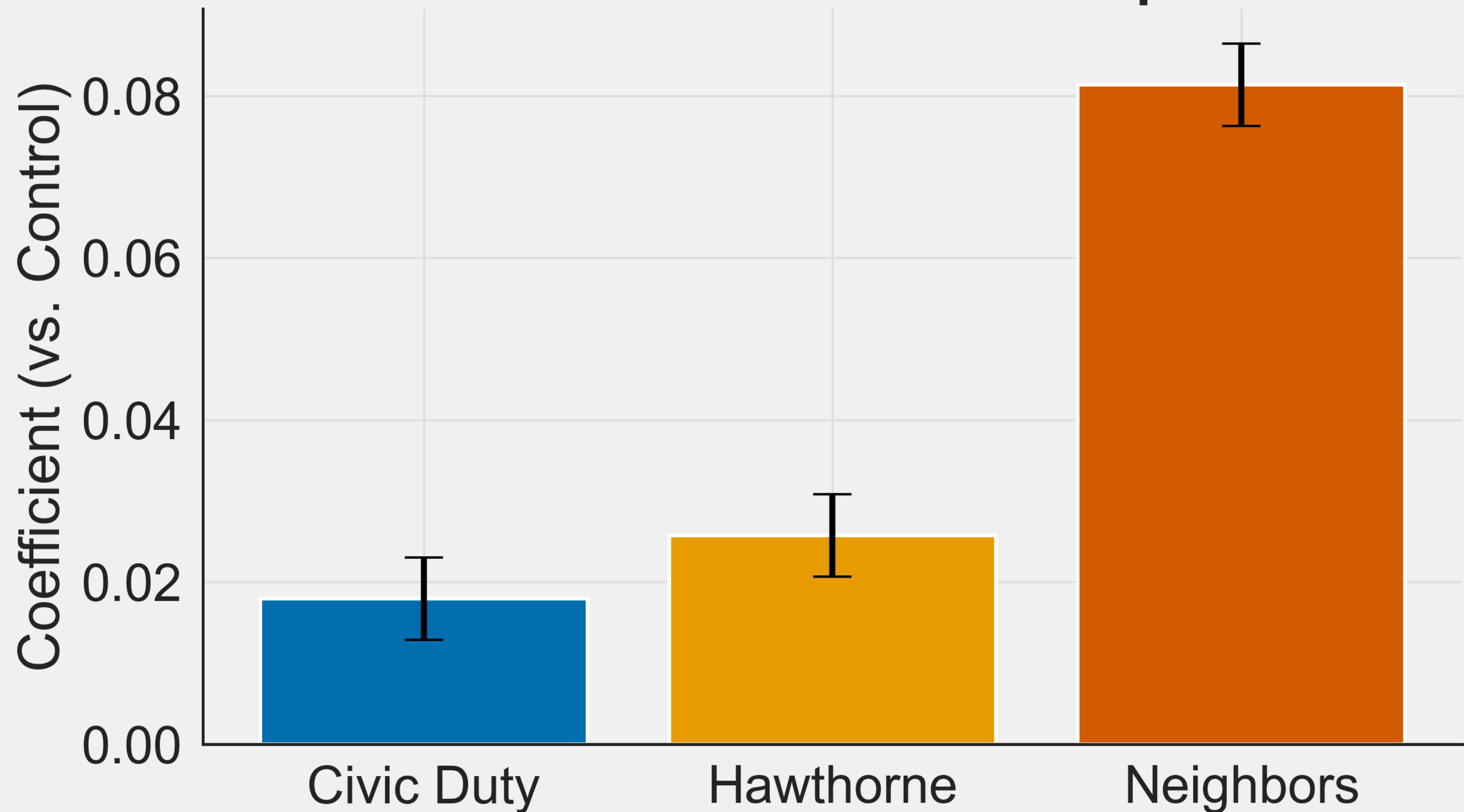
- Three dummies for four groups
- $\widehat{\beta}_0$ is the predicted value for Control ($\text{CivicDuty}_i = \text{Hawthorne}_i = \text{Neighbors}_i = 0$)

In Matrix Form

$$\mathbf{X} = \begin{pmatrix} 1 & \text{CD}_1 & \text{Hw}_1 & \text{Nb}_1 \\ 1 & \text{CD}_2 & \text{Hw}_2 & \text{Nb}_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \text{CD}_n & \text{Hw}_n & \text{Nb}_n \end{pmatrix}$$

Dummy Coefficients Are Differences

**Dummy Coefficients Are Differences
From the Baseline Group**



The Dummy Variable Trap

- What if we included a dummy for **every** level, plus the intercept?
- Every observation's dummies would sum to exactly 1 (every row belongs to exactly one group)
 - When all dummy variables equal zero, what group is the intercept β_0 representing?

Interaction Terms

Why Interactions?

- So far every predictor's effect has been assumed **constant** across the whole sample
- What if the effect of X_1 on Y depends on the value of X_2 ?
- An **interaction term** lets a predictor's effect vary by group, or by the level of another variable

The Interaction Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}) + \epsilon_i$$

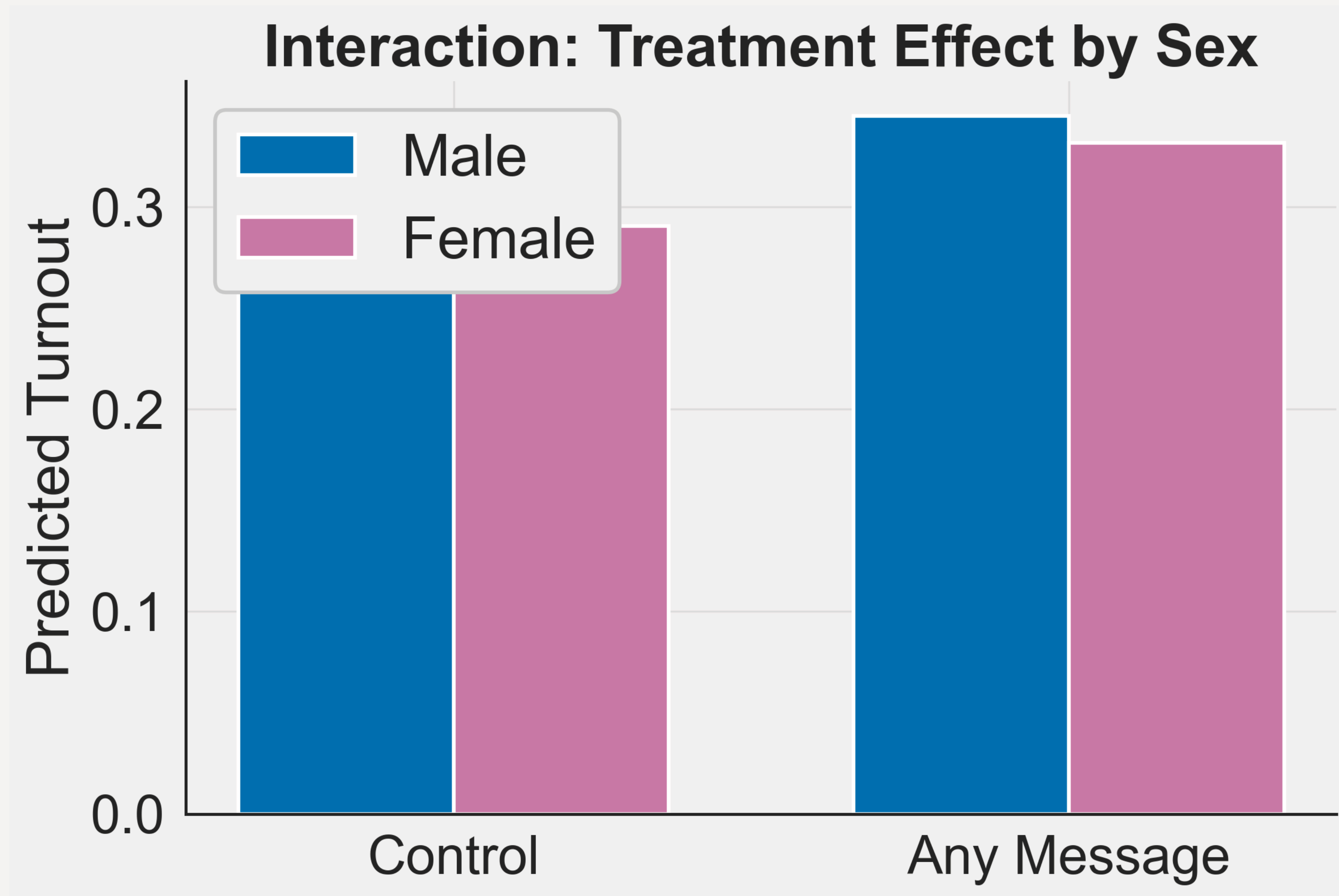
- β_3 : how much the effect of X_1 changes for a one-unit increase in X_2 (and vice versa)
- The conditional effect of X_1 on Y is $\frac{\partial Y}{\partial X} = \beta_1 + \beta_3 Z$ where β_1 is the baseline effect of X_1 when $X_2 = 0$ and β_3 is the slope change in X_1 for a one unit change in X_2 .

In Matrix Form

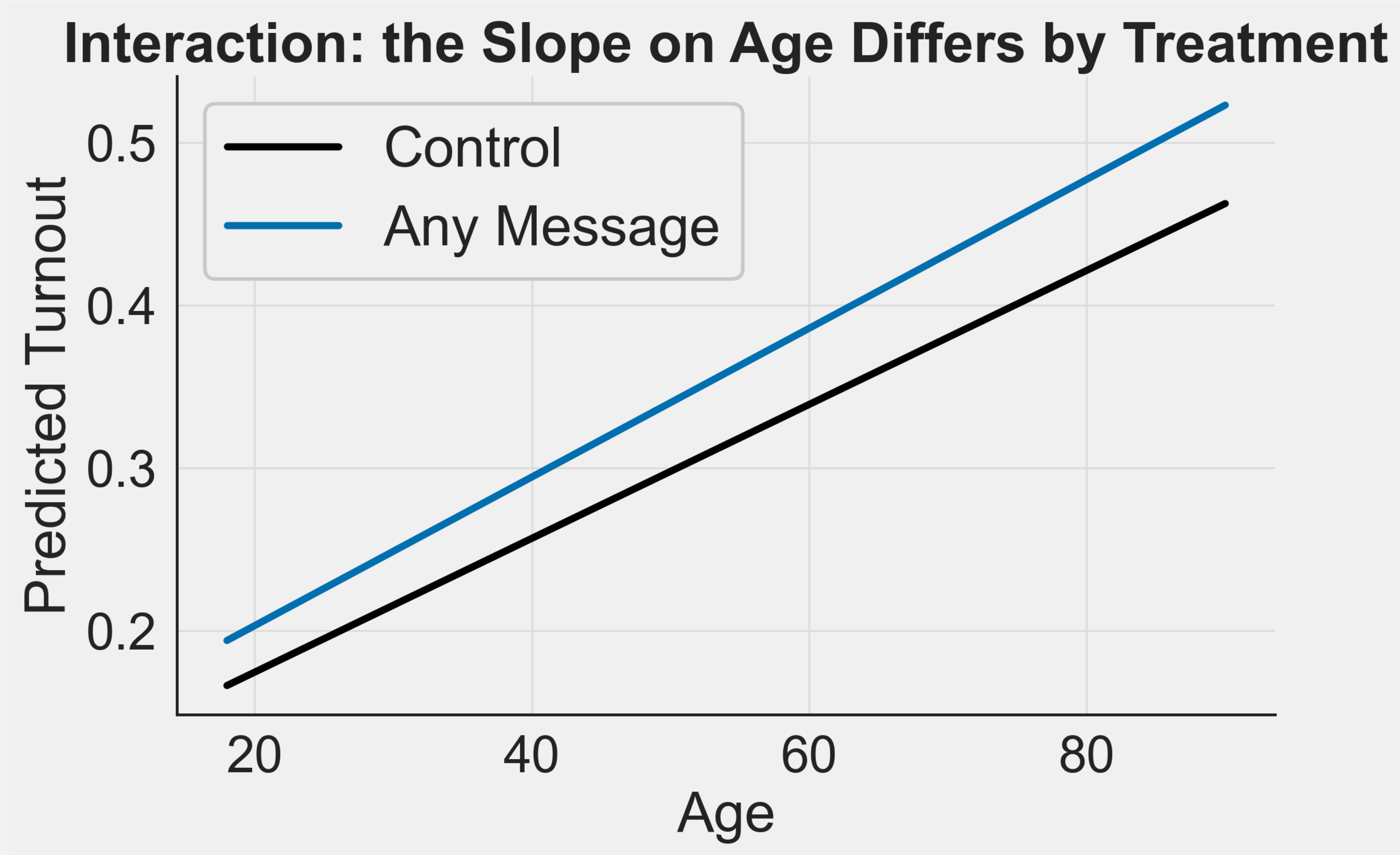
The design matrix gets one more column – the elementwise product of two existing columns:

$$\mathbf{X} = \begin{pmatrix} 1 & X_{11} & X_{21} & X_{11}X_{21} \\ 1 & X_{12} & X_{22} & X_{12}X_{22} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & X_{1n} & X_{2n} & X_{1n}X_{2n} \end{pmatrix}$$

Dummy x Dummy: Treatment Effect by Sex



Continuous x Dummy: Treatment Effect by Age



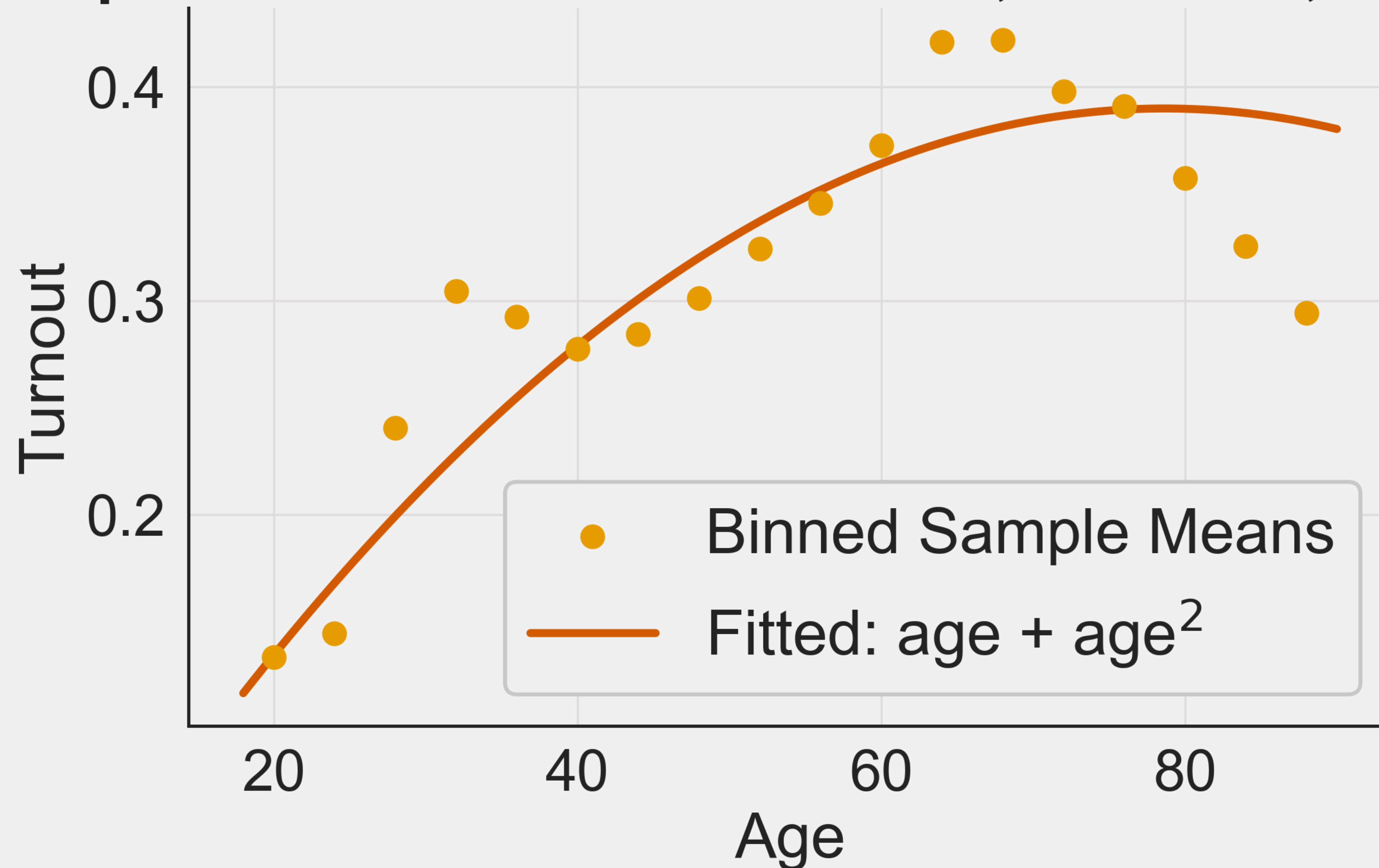
Squared Terms: a Self-Interaction

$$Y_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{age}_i^2 + \epsilon_i$$

- A squared term is X interacted with itself
- Lets the effect of X change as X itself changes, instead of forcing a single constant slope

Age and Turnout Aren't Linear

A Squared Term Lets Turnout Rise, Then Fall, With Age



Interpreting Predicted-Value Plots

- With interactions, coefficients alone get hard to read quickly
- **Predicted-value plots** (like the last several slides) show $\hat{Y}$ across levels of one predictor, with separate lines/bars for the other
- Parallel lines/bars $\Rightarrow$ no interaction. Non-parallel $\Rightarrow$ the effect of one variable depends on the other

Summary

- Multiple regression: same model, same estimation logic as simple regression – $\mathbf{X}$ just grows a column per predictor
- Omitting a confounder biases the coefficient on any predictor correlated with it; adding the confounder “holds it constant”
- A categorical variable with k levels needs $k - 1$ dummies plus the intercept – never k , or you hit the dummy variable trap
- An interaction term is just another design-matrix column: the elementwise product of two existing ones
- Predicted-value plots are the clearest way to communicate what an interaction means

Key Definitions

- **Multiple regression model:** extends the bivariate model to k predictors – $Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \epsilon_i$
- **Design matrix (X):** one row per observation, one column per predictor, plus a leading column of 1's for the intercept
- **Confounding variable:** a variable with a causal effect on both the independent and dependent variable
- **Bias:** a systematic tendency for an estimator to miss the true value, in the same direction, on average – $\text{Bias}(\hat{\beta}) = E[\hat{\beta}] - \beta$
- **Omitted variable bias:** bias caused by leaving a confounding variable out of the model

Key Definitions

- **Partial derivative** (as “holding constant”): $\frac{\partial \hat{Y}}{\partial X_j} = \hat{\beta}_j$ – treating every other predictor as fixed while differentiating
- **Dummy variable** (indicator / binary variable): a variable that takes only 0 or 1, marking group membership
- **Baseline / reference group**: the group represented by the intercept – the one where every dummy equals 0
- **Categorical variable**: a variable with k discrete levels, represented using $k - 1$ dummy variables
- **Dummy variable trap**: including a dummy for every level plus the intercept, which makes $\mathbf{X}^\top \mathbf{X}$ singular (no unique $\hat{\beta}$)

Key Definitions

- **Interaction term:** a design-matrix column formed by multiplying two predictors together, letting one predictor's effect on Y vary with the other
- **Predicted-value plot:** a plot of $\hat{Y}$ across levels of one predictor, with separate lines/bars for another – used to visualize and communicate interactions